

Mathematical Analysis II

Week 6 Homework (April 9)

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Section 9.5

2. 求下列函数由点 (x_0, y_0) 变到 $(x_0 + h, y_0 + k)$ 时函数的增量.

(1) $f(x, y) = x^3 + y^2 - 6xy - 39x + 18y + 4$, $(x_0, y_0) = (5, 6)$;

解. 计算 $f(5 + h, 6 + k)$:

$$\begin{aligned} f(5 + h, 6 + k) &= (5 + h)^3 + (6 + k)^2 - 6(5 + h)(6 + k) - 39(5 + h) + 18(6 + k) + 4 \\ &= h^3 + 15h^2 + k^2 - 6hk - 102 \end{aligned}$$

$$\text{又 } f(5, 6) = 125 + 36 - 180 - 195 + 108 + 4 = -102$$

$$\text{因此 } f(5 + h, 6 + k) - f(5, 6) = h^3 + 15h^2 + k^2 - 6hk$$

(2) $f(x, y) = x^2y + xy^2 - 2xy$, $(x_0, y_0) = (1, -1)$.

解. 计算 $f(1 + h, k - 1)$:

$$\begin{aligned} f(1 + h, k - 1) &= (1 + h)^2(k - 1) + (1 + h)(k - 1)^2 - 2(1 + h)(k - 1) \\ &= kh^2 - h^2 + k^2 - 3k + hk^2 + h + 2 - 2hk \end{aligned}$$

$$\text{又 } f(1, -1) = -1 + 1 + 2 = 2$$

$$\text{因此 } f(1 + h, k - 1) - f(1, -1) = kh^2 - h^2 + k^2 - 3k + hk^2 + h - 2hk$$

3. 对于函数 $f(x, y) = \sin \pi x + \cos \pi y$, 用中值定理证明, 存在一个数 θ , $0 < \theta < 1$ 使得

$$\frac{4}{\pi} = \cos \frac{\pi\theta}{2} + \sin \left[\frac{\pi}{2}(1 - \theta) \right]$$

证. 即证 $2 = \frac{\pi}{2} \left[\cos \frac{\pi\theta}{2} + \sin \left[\frac{\pi}{2}(1 - \theta) \right] \right]$

由二元函数中值定理, 存在 $0 < \theta < 1$ 使得

$$f(x, y) - f(x_0, y_0) = f_x(x_0 + \theta(x - x_0), y_0 + \theta(y - y_0))(x - x_0) + f_y(x_0 + \theta(x - x_0), y_0 + \theta(y - y_0))(y - y_0)$$

对 f 求偏导, 得到

$$\begin{cases} f_x = \pi \cos \pi x \\ f_y = -\pi \sin \pi y \end{cases}$$

取 $(x_0, y_0) = \left(0, \frac{1}{2}\right)$, $(x, y) = \left(\frac{1}{2}, 0\right)$, 则 $(x - x_0, y - y_0) = \left(\frac{1}{2}, -\frac{1}{2}\right)$

从而

$$\begin{cases} x_0 + \theta(x - x_0) = \frac{\theta}{2} \\ y_0 + \theta(y - y_0) = \frac{1 - \theta}{2} \\ f\left(\frac{1}{2}, 0\right) = 2 \\ f\left(0, \frac{1}{2}\right) = 0 \end{cases}$$

代入中值定理公式, 得到

$$\begin{aligned} 2 &= \frac{1}{2}\pi \cos \frac{\pi\theta}{2} + \frac{1}{2}\pi \sin \frac{\pi(1-\theta)}{2} \\ &= \frac{\pi}{2} \left[\cos \frac{\pi\theta}{2} + \sin \left[\frac{\pi}{2}(1-\theta) \right] \right] \end{aligned}$$

两边同时乘以 $\frac{2}{\pi}$, 可得 $\frac{4}{\pi} = \cos \frac{\pi\theta}{2} + \sin \left[\frac{\pi}{2}(1-\theta) \right]$

Q.E.D.

4. 求下列函数的 Taylor 公式, 并指出展开式成立的区域.

(2) $f(x, y) = \sqrt{1 - x^2 - y^2}$ 在点 $(0, 0)$, 直到四阶为止;

解. 记 $u = x^2 + y^2$, 则 $f = \sqrt{1 - u} = 1 - \frac{1}{2}u - \frac{1}{8}u^2 + o(u^2)$

回代得到 $f = 1 - \frac{x^2 + y^2}{2} - \frac{x^4 + 2x^2y^2 + y^4}{8} + o(\rho^4)$

整理得 $f(x, y) = 1 - \frac{x^2 + y^2}{2} - \frac{x^4 + y^4}{8} - \frac{x^2y^2}{4} + o(\rho^4)$

成立区域: $x^2 + y^2 \leq 1$

(4) $\arctan \frac{1+x+y}{1-x+y}$ 在点 $(0, 0)$, 直到二阶为止;

解. $f = \arctan \frac{1 + \frac{x}{1+y}}{1 - \frac{x}{1+y}}$

记 $u = 1, v = \frac{x}{1+y}$, 则 $f = \arctan \frac{u+v}{1-uv} = \arctan u + \arctan v = \frac{\pi}{4} + \arctan \frac{x}{1+y}$, 其

中 $|uv| = \left| \frac{x}{1+y} \right| < 1$

记 $g = \arctan \frac{x}{1+y}$, 则

$$g = g(0, 0) + g_x(0, 0)x + g_y(0, 0)y + \frac{1}{2}(g_{xx}(0, 0)x^2 + 2g_{xy}(0, 0)xy + g_{yy}(0, 0)y^2) + o(\rho^2)$$

其中

$$\begin{cases} g_x = \frac{1+y}{(1+y)^2+x^2} & g_x(0,0) = 1 \\ g_y = -\frac{x}{(1+y)^2+x^2} & g_y(0,0) = 0 \end{cases}$$

从而

$$\begin{cases} g_{xx} = -\frac{2x(1+y)}{((1+y)^2+x^2)^2} & g_{xx}(0,0) = 0 \\ g_{xy} = -\frac{(1+y)^2-x^2}{((1+y)^2+x^2)^2} & g_{xy}(0,0) = -1 \\ g_{yy} = \frac{2x(y+1)}{((1+y)^2+x^2)^2} & g_{yy}(0,0) = 0 \end{cases}$$

又 $g(0,0) = 0$

因此 $g = x - xy + o(\rho^2)$

故 $f = \frac{\pi}{4} + x - xy + o(\rho^2)$

(6) $\frac{\cos x}{\cos y}$ 在点 $(0,0)$ 展开到二阶为止;

解. $\cos x = 1 - \frac{1}{2}x^2 + o(x^2)$, $\cos y = 1 - \frac{1}{2}y^2 + o(y^2)$

记 $u = \frac{1}{2}y^2$, 由于 $\frac{1}{1-u} = 1 + u + o(u^2)$, 因此

$$\frac{1}{\cos y} = \frac{1}{1 - \frac{1}{2}y^2 + o(y^2)} = 1 + u = 1 + \frac{1}{2}y^2 + o(y^2)$$

从而

$$\frac{\cos x}{\cos y} = \left(1 - \frac{1}{2}x^2\right) \left(1 + \frac{1}{2}y^2\right) + o(\rho^2) = 1 - \frac{1}{2}x^2 + \frac{1}{2}y^2 + o(\rho^2)$$

(只保留二阶项)

5. 设 $z = z(x, y)$ 是由方程 $z^3 - 2xz + y = 0$ 所确定的隐函数, 当 $x = 1, y = 1$ 时 $z = 1$, 试按 $(x-1)$ 和 $(y-1)$ 的乘幂展开函数 z 至二次项为止.

解. 对方程两边关于 x 求偏导: $3z_x z^2 - 2z - 2xz_x = 0$, 因此 $z_x = \frac{2z}{3z^2 - 2x}$, 代入 $(1, 1, 1)$ 得到 $z_x(1) = 2$

对方程两边关于 y 求偏导: $3z_y z^2 - 2xz_y + 1 = 0$, 因此 $z_y = -\frac{1}{3z^2 - 2x}$, 代入 $(1, 1, 1)$ 得到 $z_y(1) = -1$

求 z_{xx} : 对 $3z_x z^2 - 2z - 2xz_x = 0$ 两边关于 x 求偏导: $3z_{xx} z^2 + 6z_x^2 z - 2z_x - 2z_x - 2xz_{xx} = 0$, 整理得 $(3z^2 - 2x)z_{xx} = 2z_x + 2z_x - 6z_x^2 z = 8 - 24z$, 因此 $z_{xx} = \frac{8 - 24z}{3z^2 - 2x}$, 代入 $(1, 1, 1)$ 得到 $z_{xx}(1) = -16$

求 z_{xy} : 对 $3z_x z^2 - 2z - 2xz_x = 0$ 两边关于 y 求偏导: $3z_{xy} z^2 + 6z_x z_y z - 2z_y - 2xz_{xy} = 0$, 整理得 $(3z^2 - 2x)z_{xy} = 2z_y - 6z_x z_y z = -2 + 12z$, 因此 $z_{xy} = \frac{-2 + 12z}{3z^2 - 2x}$, 代入 $(1, 1, 1)$ 得到 $z_{xy}(1) = 10$

求 z_{yy} : 对 $3z_y z^2 - 2xz_y + 1 = 0$ 两边关于 y 求偏导: $3z_{yy}z^2 + 6z_y^2 z - 2xz_{yy} = 0$, 整理得 $(3z^2 - 2x)z_{yy} = -6z_y^2 z = -6z$, 因此 $z_{yy} = -\frac{6z}{3z^2 - 2x}$, 代入 $(1, 1, 1)$ 得到 $z_{yy}(1) = -6$
因此

$$\begin{aligned} z(x, y) &= z(1, 1) + z_x(x-1) + z_y(y-1) \\ &\quad + \frac{1}{2} (z_{xx}(x-1)^2 + 2z_{xy}(x-1)(y-1) + z_{yy}(y-1)^2) + o(\rho^2) \\ &= 1 + 2(x-1) - (y-1) - 8(x-1)^2 + 10(x-1)(y-1) - 3(y-1)^2 + o(\rho^2) \end{aligned}$$

6. 证明, 球面三角学中的余弦定理

$$\cos z = \cos x \cos y + \sin x \sin y \cos \theta$$

在原点的邻域内转化为欧几里得几何中的余弦定理

$$z^2 = x^2 + y^2 - 2xy \cos \theta$$

证. 将 $\cos x, \cos y, \cos z, \sin x, \sin y$ 展开至二阶, 可得

$$\begin{cases} \cos x \approx 1 - \frac{x^2}{2} \\ \cos y \approx 1 - \frac{y^2}{2} \\ \cos z \approx 1 - \frac{z^2}{2} \\ \sin x \approx x \\ \sin y \approx y \end{cases}$$

代入原式, 得到 $1 - \frac{z^2}{2} = \left(1 - \frac{x^2}{2}\right) \left(1 - \frac{y^2}{2}\right) + xy \cos \theta$

忽略高阶无穷小项, 整理可得 $z^2 = x^2 + y^2 - 2xy \cos \theta$

Q.E.D.

7. 求下列函数的极值.

(1) $f(x, y) = xy + \frac{50}{x} + \frac{20}{y} \quad (x > 0, y > 0);$

解. 由极值的必要条件得 $f_x = f_y = 0$

求得 $f_x = y - \frac{50}{x^2} = 0$, 即 $y = \frac{50}{x^2}$

$f_y = x - \frac{20}{y^2} = 0$, 即 $x = \frac{20}{y^2}$

联立方程组, 解得

$$\begin{cases} x = 5 \\ y = 2 \end{cases}$$

计算二阶偏导数: $A = f_{xx} = \frac{100}{x^3} = \frac{4}{5} > 0$

$$B = f_{xy} = 1$$

$$C = f_{yy} = \frac{40}{y^3} = 5$$

$$\text{因此 } \Delta = AC - B^2 = 3 > 0$$

由于 $A > 0$ 且 $\Delta > 0$, 因此该点为极小值点, 极小值为 $f(5, 2) = 30$

$$(3) f(x, y) = e^{2x}(x + 2y + y^2);$$

解. 求得 $f_x = 2e^{2x}(x + 2y + y^2) + e^{2x} = e^{2x}(2x + 4y + 2y^2 + 1) = 0$

$$f_y = e^{2x}(2 + 2y) = 0$$

由于 $e^{2x} > 0$, 联立方程组, 解得

$$\begin{cases} x = \frac{1}{2} \\ y = -1 \end{cases}$$

$$\text{计算二阶偏导数: } A = f_{xx} = 4e^{2x}(x + 2y + y^2) + 4e^{2x} = 4e^{2x}(x + 2y + y^2 + 1) = 2e > 0$$

$$B = f_{xy} = e^{2x}(4 + 4y) = e(4 - 4) = 0$$

$$C = f_{yy} = 2e^{2x} = 2e$$

$$\text{因此 } \Delta = AC - B^2 = 4e^2 > 0$$

由于 $A > 0$ 且 $\Delta > 0$, 因此该点为极小值点, 极小值为 $f\left(\frac{1}{2}, -1\right) = e\left(\frac{1}{2} - 2 + 1\right) = -\frac{e}{2}$

$$(5) x^2 + y^2 + z^2 - 2x + 2y - 4z - 10 = 0, \text{ 求隐函数 } z = z(x, y) \text{ 的极值.}$$

解. 记 $F(x, y, z) = x^2 + y^2 + z^2 - 2x + 2y - 4z - 10 = 0$

由极值的必要条件得 $z_x = -\frac{F_x}{F_z} = z_y = -\frac{F_y}{F_z} = 0$, 即 $F_x = F_y = 0$ 且 $F_z \neq 0$

求得 $F_x = 2x - 2 = 0$, 即 $x = 1$

$F_y = 2y + 2 = 0$, 即 $y = -1$

代入原方程, 得到 $1 + 1 + z^2 - 2 - 2 - 4z - 10 = 0$, 即 $z^2 - 4z - 12 = 0$

解得 $(z - 6)(z + 2) = 0$, 因此 $z = 6$ 或 $z = -2$

计算二阶偏导数: 对方程两边关于 x 求偏导, 得到 $z_{xx}(z - 2) + (z_x)^2 = -1$; 关于 y 求偏

导: $z_{yy}(z - 2) + (z_y)^2 = -1$; 混合偏导: $z_{xy}(z - 2) + z_x z_y = 0$

由于 $z_x = z_y = 0$, 因此

$$\begin{cases} z_{xx} = \frac{-1}{z - 2} \\ z_{yy} = \frac{-1}{z - 2} \\ z_{xy} = 0 \end{cases}$$

• 当 $z = 6$ 时,

$$\begin{cases} A = z_{xx} = -\frac{1}{4} < 0 \\ B = z_{xy} = 0 \\ C = z_{yy} = -\frac{1}{4} \end{cases}$$

因此 $\Delta = AC - B^2 = \frac{1}{16} > 0$

由于 $A < 0$ 且 $\Delta > 0$, 因此该点为极大值点, 极大值为 $z = 6$

• 当 $z = -2$ 时,

$$\begin{cases} A = z_{xx} = \frac{1}{4} > 0 \\ B = z_{xy} = 0 \\ C = z_{yy} = \frac{1}{4} \end{cases}$$

因此 $\Delta = AC - B^2 = \frac{1}{16} > 0$

由于 $A > 0$ 且 $\Delta > 0$, 因此该点为极小值点, 极小值为 $z = -2$

第 9 章综合习题

4. 设 $f(x, y, z)$ 是 n 次齐次的可微函数. 若方程 $f(x, y, z) = 0$ 隐含函数 $z = \varphi(x, y)$ (即 $\frac{\partial f}{\partial z} \neq 0$), 则 $\varphi(x, y)$ 是一次齐次函数.

证. 由隐函数定义得 $f(x, y, \varphi(x, y)) = 0$

由于 f 是 n 次齐次可微函数, 因此

$$f(tx, ty, t\varphi(x, y)) = t^n f(x, y, \varphi(x, y)) = 0$$

从而 $z = t\varphi(x, y)$ 是方程 $f(tx, ty, z) = 0$ 的一个解

又方程 $f(tx, ty, z) = 0$ 隐含函数 $z = \varphi(tx, ty)$. 由于 $\frac{\partial f}{\partial z} \neq 0$, 因此 $\varphi(tx, ty) = t\varphi(x, y)$
故 $\varphi(x, y)$ 是一次齐次函数

Q.E.D.

5. 设 $f(x, y)$ 在 $\mathbb{R}^2$ 上有连续二阶偏导数, 且对任意实数 x, y, z 满足 $f(x, y) = f(y, x)$ 和

$$f(x, y) + f(y, z) + f(z, x) = 3f\left(\frac{x+y+z}{3}, \frac{x+y+z}{3}\right).$$

试求 $f(x, y)$.

解. 记 $g(u) = f(u, u)$

$$\text{因此 } f(x, y) + f(y, z) + f(z, x) = 3g\left(\frac{x+y+z}{3}\right)$$

$$\text{关于 } x \text{ 求偏导: } \frac{\partial f}{\partial x}(x, y) + \frac{\partial f}{\partial y}(z, x) = g'\left(\frac{x+y+z}{3}\right)$$

$$\text{关于 } z \text{ 求偏导: } \frac{\partial f}{\partial y}(y, z) + \frac{\partial f}{\partial x}(z, x) = g'\left(\frac{x+y+z}{3}\right)$$

又由于 $f(x, y) = f(y, x)$, 因此 $\frac{\partial f}{\partial x}(x, y) = \frac{\partial f}{\partial x}(y, x)$, 从而

$$\frac{\partial f}{\partial x}(x, y) + \frac{\partial f}{\partial x}(x, z) = \frac{\partial f}{\partial x}(z, y) + \frac{\partial f}{\partial x}(z, x)$$

关于 y 求偏导: $\frac{\partial^2 f}{\partial x \partial y}(x, y) = \frac{\partial^2 f}{\partial x \partial y}(z, y)$

因此 $\frac{\partial^2 f}{\partial x \partial y}$ 与第一个变量无关, 由对称性同理可得与第二个变量无关。

不妨设 $\frac{\partial^2 f}{\partial x \partial y} = A$ 为常数

因此 $f(x, y) = Axy + B(x) + B(y)$

代入原方程得

$$Axy + Ayz + Azx + 2B(x) + 2B(y) + 2B(z) = \frac{A}{3}(x + y + z)^2 + 6B\left(\frac{x + y + z}{3}\right)$$

整理得

$$\frac{Axy + Ayz + Azx}{3} + 2B(x) + 2B(y) + 2B(z) = \frac{A}{3}(x^2 + y^2 + z^2) + 6B\left(\frac{x + y + z}{3}\right)$$

由轮换对称性, 可得

$$\frac{Axy}{3} + B(x) + B(y) = \frac{A}{9}(x^2 + y^2 + z^2) + 2B\left(\frac{x + y + z}{3}\right)$$

因此 $B(x) = \frac{A}{4}x^2 + bx + c$, 记 $a = \frac{A}{4}$

故 $f(x, y) = a(x^2 + 4xy + y^2) + b(x + y) + c$

6. 证明不等式 $\frac{x^2 + y^2}{4} \leq e^{x+y-2}$ ($x \geq 0, y \geq 0$).

证. 由于 $e^{t-1} \geq t$, 取 $t = \frac{x+y}{2}$, 可得 $e^{\frac{x+y}{2}-1} \geq \frac{x+y}{2} \geq 0$

因此

$$e^{x+y-2} = \left(e^{\frac{x+y}{2}-1}\right)^2 \geq \left(\frac{x+y}{2}\right)^2 = \frac{x^2 + 2xy + y^2}{4} \geq \frac{x^2 + y^2}{4}$$

Q.E.D.

7. 设在 $\mathbb{R}^3$ 上定义的 $u = f(x, y, z)$ 是 z 的连续函数, 且 $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ 在 $\mathbb{R}^3$ 上连续. 证明 u 在 $\mathbb{R}^3$ 上连续.

证. 对任意点 $P_0(x_0, y_0, z_0)$, 需证明:

对任意 $\varepsilon > 0$,

存在 $\delta > 0$,

当 $\sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} < \delta$ 时,

有 $|f(x, y, z) - f(x_0, y_0, z_0)| < \varepsilon$.

由三角不等式得

$$|f(x, y, z) - f(x_0, y_0, z_0)| \leq |f(x, y, z) - f(x_0, y_0, z)| + |f(x_0, y_0, z) - f(x_0, y_0, z_0)|$$

固定 z , 在 xy 平面上存在 ξ, η 使得

$$f(x, y, z) - f(x_0, y_0, z) = \frac{\partial f}{\partial x}(\xi, \eta, z)(x - x_0) + \frac{\partial f}{\partial y}(\xi, \eta, z)(y - y_0)$$

由于 $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ 在 $\mathbb{R}^3$ 上连续, f 在 P_0 的某个闭邻域内必定是有界的。设界为 M , 则

$$|f(x, y, z) - f(x_0, y_0, z)| \leq M|x - x_0| + M|y - y_0|$$

当 $|x - x_0| < \frac{\varepsilon}{4M}$, $|y - y_0| < \frac{\varepsilon}{4M}$ 时, $|f(x, y, z) - f(x_0, y_0, z)| < \frac{\varepsilon}{2}$ 。

由于 f 关于 z 连续, 固定 (x_0, y_0) , 当 $z \rightarrow z_0$ 时, $|f(x_0, y_0, z) - f(x_0, y_0, z_0)| \rightarrow 0$ 。即存在 $\delta_1 > 0$, 当 $|z - z_0| < \delta_1$ 时, $|f(x_0, y_0, z) - f(x_0, y_0, z_0)| < \frac{\varepsilon}{2}$ 。

因此, 对任意 $\varepsilon > 0$, 存在 $\delta = \min\left(1, \frac{\varepsilon}{4M}, \delta_1\right) > 0$, 当 $\sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} < \delta$ 时, 有 $|f(x, y, z) - f(x_0, y_0, z_0)| < \varepsilon$, 故 u 在 $\mathbb{R}^3$ 上全局连续。

Q.E.D.